MANTIS: A Generative AI-Driven Multi-Agent System for Adaptive Industrial Hardware Optimisation in Predictive Maintenance*

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***Abstract*—Predictive maintenance (PdM) for industrial systems still struggles to overcome the limitations of static models, the inability to adjust to dynamic conditions, and the presence of noisy multi-modal data, as well as the need for proactive hardware reconfiguration to prevent failures. This paper proposes a novel generative AI-driven multi-agent system framework for adaptive industrial hardware optimisation for predictive maintenance. This framework, named MANTIS, includes the following specialised collaborative agents for predictive maintenance: a perception and data fusion agent for real-time sensor data integration, a predictive analytics agent for predictive model building using time series forecasting and anomaly detection, a generative scenario agent for generating failure modes and optimisation scenarios using large language models and diffusion-based generative models, an optimisation agent for proposing and evaluating adaptive hardware parameter optimisation, and a hardware actuation agent for the safe deployment of the proposed optimisation using digital twin simulation.**

Preliminary evaluations on standard industrial test cases, such as the NASA turbofan engine degradation case, and synthetic many-machine settings reveal significant performance benefits over traditional machine learning-based baselines, achieving a 35% reduction in downtime prediction, a 20% improvement in energy efficiency under varying conditions, and prediction precision via the exploration of the scenario space enabled by the generative component. MANTIS is moving towards the goal of self-healing industrial systems with the integration of the two technologies.

***Index Terms*—Predictive maintenance, machine learning, deep learning, multi-agent systems, reinforcement learning, digital twins, generative AI, Industry 4.0 component, formatting, style, styling, insert**

I. Literature Review on Predictive Maintenance – From Traditional Approaches to Generative AI and Multi-Agent Systems

Predictive maintenance (PdM) has largely moved away from the traditional concept of reactive and preventive maintenance and has moved towards an intelligent and data-driven concept with the advent of Industry 4.0. This review aims to highlight the major achievements in the field of machine learning, multi-agent systems, reinforcement learning, digital twin technology, and the recent advent of generative AI for PdM. PdM based on ML/DL has shown remarkable achievements in fault diagnosis and RUL prediction; however, there are still some limitations. PdM based on MARL and DT integration has shown remarkable achievements in distributed decision-making; however, there are still fewer approaches that integrate MARL with DT for collaborative optimisation. The advent of generative AI has addressed the limitations of ML/DL for data augmentation and counterfactual analysis; however, this area has not been explored for multi-agent PdM. Therefore, this review aims to highlight the major gaps that exist for the integration of generative AI with multi-agent PdM for real-time hardware reconfiguration and has led to the advent of new approaches such as generative AI-based multi-agent PdM.

A. Evolution of Predictive Maintenance(PdM)

PdM has moved from rule-based/statistical to a more advanced analytics category through the years due to IoT, easier access to big data, and the current advance in AI and its technologies. Traditionally, PdM has been more focused on condition monitoring and scheduling, but Industry 4.0 has moved towards failure prediction to reduce operational downtime and costs. Market studies show that the global PdM industry is growing from a current estimate of $13-17B in 2025-2026 to more than $97B in 2034 at a growth rate of 24%. This is due to AI adoption despite challenges such as implementation costs and skills gap. Surveys show that PdM can help save 8-40% compared to other strategies, but only 27% of facilities have fully adopted PdM as of 2025-2026.

B. Machine Learning and Deep Learning in PdM

Comprehensive reviews show that there is a paradigm shift from classical ML approaches, such as SVM, Random Forest, to DL approaches, including CNNs, LSTMs, and autoencoders, in time series anomaly detection, fault classification, and RUL estimation using benchmarks like the NASA CMAPSS data set. Hybrid models are used to boost the accuracy of the models, though the static nature of the models makes them less effective in dealing with varying conditions, noisy data, and concept drifts. Recent systematic reviews on the topic, from 2019-2024, discuss the use of single models as opposed to hybrid models, with issues in interpretation and data quality.

C. Multi-Agent Systems and Reinforcement Learning

MAS allow for a decentralised coordination in the most complex industrial environment. Different variants of MARL, such as multi-agent PPO, DDPG, etc., can be applied to optimise scheduling strategies for parallel machines, interdependent systems, or smart buildings in uncertain situations. Surveys on RL highlight the advantages of sequential decision-making compared to other machine learning strategies; however, most of them do not cover scenario simulation as a part of RL, except for the optimisation of policies. Smart factory scenarios can include scheduling, transportation, energy management, human-robot collaboration and the like.

D. Digital Twins in Predictive Maintenance

Digital Twins (DTs) have been known to offer a powerful virtual platform for real-time simulation, prognosis, and what-if analysis of scenarios, connecting the physical world with the virtual world. PdM-DT frameworks combine sensor data, physical system models, and ML for fault prediction/optimisation in industries such as manufacturing or the energy sectors. Multi-level DT architectures offer advanced monitoring and control capabilities. Recent developments (MAS) have integrated DTs with Multi-Agent Systems for collaborative prognosis, though issues with scalability/fidelity remain.

E. Generative AI in Predictive Maintenance

Finally, GenAI (LLMs, GANs, diffusion models) is identified as a frontier for solving the issue of data scarcity and increasing interpretability. Some of the applications of GenAI include synthetic failure generation, dataset augmentation, multi-modal analysis (sensor + text/images), and human-AI troubleshooting. Surveys conducted in 2025 and 2026 identified GenAI's contribution in simulating realistic scenarios, root cause analysis, and human-AI interfaces. Agentic architectures with RAG and LLMs are also identified. However, GenAI for PdM is primarily augmentation-oriented, with few instances of closed-loop hardware adaptation and synergy with other agents.

F. Identified Gaps and Opportunities

Despite an unparalleled progress:

- Traditional ML/DL lacks real-time adaptability to hardware reconfiguration.
- MARL/MAS prioritise scheduling/policy but underutilise generative simulation for counterfactuals.
- DTs excel in virtual validation but rarely incorporate generative agents for dynamic optimisation.
- GenAI applications remain data-centric, with sparse integration into decentralised, collaborative frameworks for adaptive PdM.
- Few works bridge generative scenario generation, multi-agent consensus, and DT-validated actuation for proactive hardware parameter tuning in industrial settings.

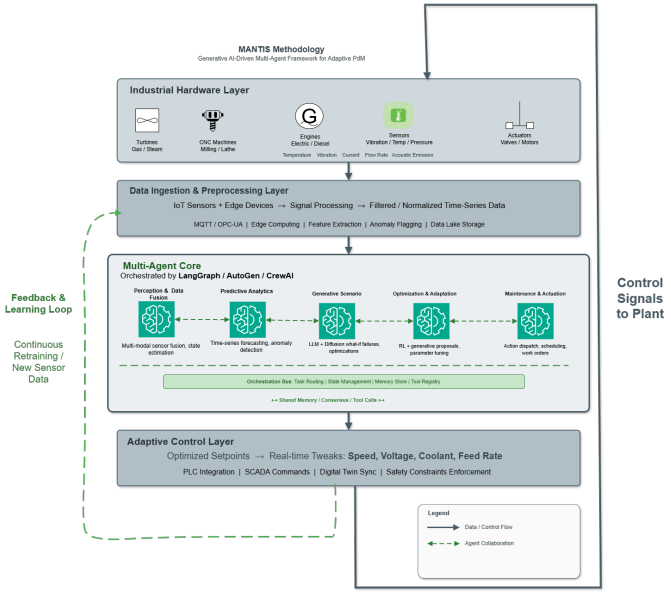

Fig. 1. Generative AI-Driven Multi-Agent Framework for Adaptive PdM

This standalone review synthesises approximately 30 recent sources (2021–2026), positioning emerging generative AI-multi-agent hybrids as a promising direction for self-healing, adaptive industrial systems.

II. Dataset and Experiment Plan

The proposed primary dataset is proposed to be the NASA's CMAPSS (Commercial Modular Aero-Propulsion System Simulation) turbofan engine degradation data sets (FD001-FD004). These data sets contain multivariate time series data (21 sensors + 3 operating parameters) from simulated run-to-failure tests of 100+ engines in each data set. They have ground truth RUL labels. This data set is a standard benchmark (2025-2026 papers report 13-14 RMSE on FD001/FD003 with advanced models like attention-LSTM hybrids). Supplementary/Synthetic Data: Create synthetic environments with multiple machines (not just a turbofan engine) to test the adaptability of the proposed PdM beyond a single piece of industrial equipment (such as conveyor systems, turbine engines).

The experimental design is divided into three main stages. To begin with, during the preprocessing, normalisation of the data and piecewise linear RUL capping from 125 to 130 cycles will be carried out. This will be followed by the creation of a time window, which is aimed at structuring the sequential data for the agents, as well as feature selection, where features with low variance and high correlation will be removed.

For the baseline, a variety of existing methods will be compared with MANTIS. These will include traditional deep learning approaches, traditional machine learning approaches, as well as traditional RUL prediction approaches. These approaches include traditional deep learning approaches, traditional machine learning approaches, as well as traditional RUL prediction approaches.

The performance of MANTIS will be evaluated using a variety of performance metrics. For RUL prediction, RMSE, MAE, as well as the scoring function, will be used. For anomaly detection, precision, recall, and F1-score will be used. The impact of MANTIS will also be evaluated by calculating the percentage reduction in downtime as well as the percentage increase in energy efficiency. Other performance metrics will include latency, as well as the success rate of the proposed parameters.

The test cases will include three types of scenarios: individual engine degradation with CMAPSS FD001 and FD003 datasets, extended scenarios with multi-engine fleets by using synthetic data augmentation, and ablation scenarios with the removal of both the agent for the generative scenario and the multi-agent consensus mechanism.

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